

SIMATS ENGINEERING

ASSESSMENT TOOL 1:

Experiments

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE OUTCOME COVERED

COURSE CODE: CSA1508

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3) Question Count: 4

Code of Conduct

Total Marks: 40

I, **G. Umesh Chandra Yadav/192525424** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: **G. Umesh**

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis

Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation
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Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Experiment 1: Create a Cloud-based Hospital Information System (SaaS)

Aim: To develop a cloud-based Hospital Information System using a Cloud Service Provider to store and manage patient records online, thereby demonstrating the Software as a Service (SaaS) cloud model.

Tools / Platform

- Google App Sheet
- Google Sheets
- Google Drive
- Zoho creator
- Internet Connection

Procedure

1. Create a Google Sheet with fields such as Patient Name, Age, Address, Phone Number, Email, Attender Name, Attender Phone, Doctor Name, Disease, and Date of Visit.
2. Sign in to Google AppSheet using a Google account.
3. Select **Create App** and import the Google Sheet as the data source.
4. Customize the patient registration form and configure field validations.
5. Create views for patient records and search functionality.
6. Test the application by entering sample patient details.
7. Verify that the entered data is automatically synchronized with Google Sheets.
8. Publish and access the application through a web browser or mobile device.

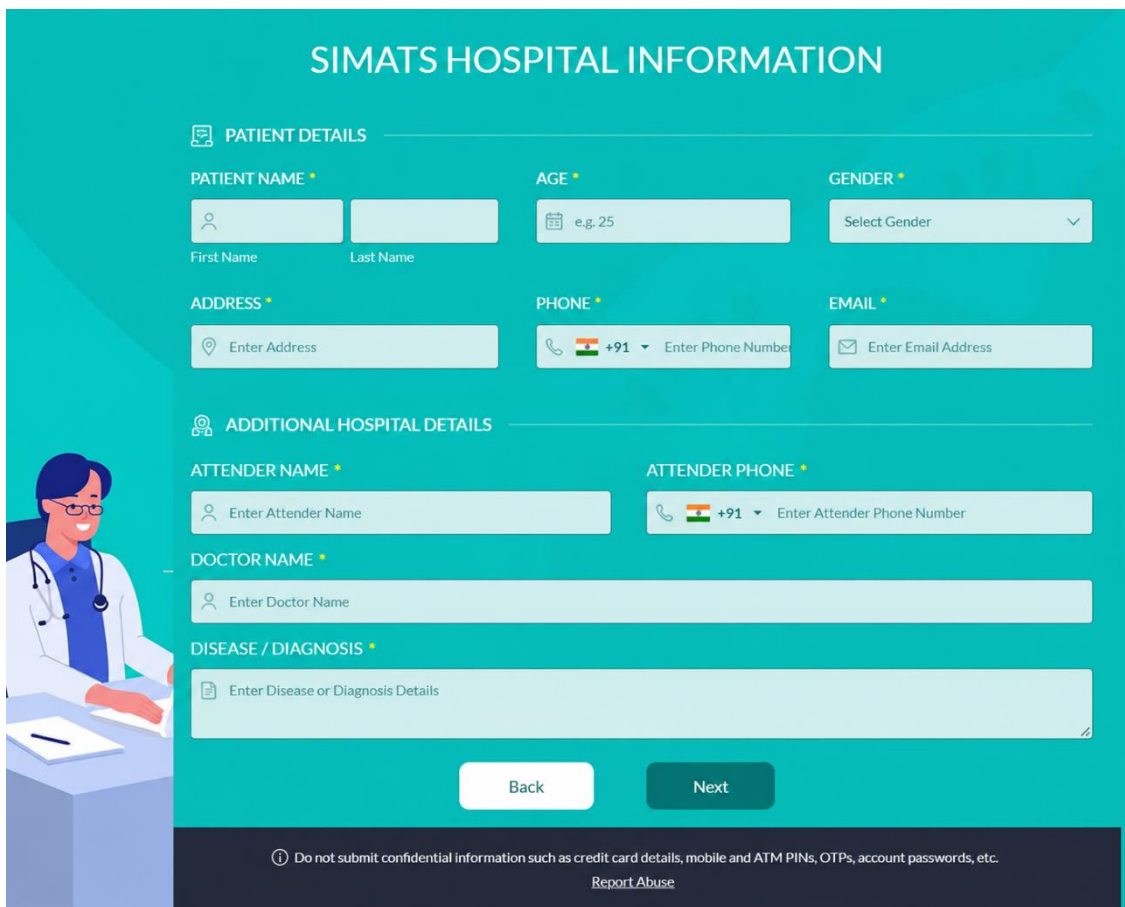
Application of Concepts

- Demonstrated the **SaaS cloud model** using Google AppSheet.

- Stored hospital records in cloud storage instead of local storage.
- Implemented real-time synchronization between the application and cloud database.
- Enabled centralized management of patient records.
- Allowed secure access from multiple devices through the internet.

Analysis and Output:

The Hospital Information System successfully stored patient details such as patient name, age, address, phone number, email, attender information, doctor name, and disease. Data entered into the application was immediately synchronized with the cloud database. The application demonstrated easy accessibility, centralized data management, improved efficiency, and reduced paperwork.



The image shows a web-based form titled "SIMATS HOSPITAL INFORMATION". The form is divided into two main sections: "PATIENT DETAILS" and "ADDITIONAL HOSPITAL DETAILS".

PATIENT DETAILS

- PATIENT NAME ***: Two input fields for "First Name" and "Last Name".
- AGE ***: An input field with a calendar icon and placeholder text "e.g. 25".
- GENDER ***: A dropdown menu labeled "Select Gender".
- ADDRESS ***: An input field with a location pin icon and placeholder text "Enter Address".
- PHONE ***: An input field with a phone icon, a dropdown for country code "+91", and placeholder text "Enter Phone Number".
- EMAIL ***: An input field with an email icon and placeholder text "Enter Email Address".

ADDITIONAL HOSPITAL DETAILS

- ATTENDER NAME ***: An input field with a person icon and placeholder text "Enter Attender Name".
- ATTENDER PHONE ***: An input field with a phone icon, a dropdown for country code "+91", and placeholder text "Enter Attender Phone Number".
- DOCTOR NAME ***: An input field with a person icon and placeholder text "Enter Doctor Name".
- DISEASE / DIAGNOSIS ***: A large text area with a document icon and placeholder text "Enter Disease or Diagnosis Details".

At the bottom of the form, there are two buttons: "Back" and "Next".

Below the form, there is a dark blue footer bar containing a warning icon and text: "Do not submit confidential information such as credit card details, mobile and ATM PINs, OTPs, account passwords, etc." and a link "Report Abuse".

Patient Registration Report

+ Add

	Patient Name	Gender	Address	Phone Number	Email	Attender Name	Attender Phone	Doctor Name	Disease	Date Of Visit	Patient Code
	Michael Clark	Female	321 Hill Street	+18319801245	mclark@email.com	Amy Johnson	+17127189724	Dr. Kevin Brown	Cold	06-Aug-2026	P005
	Patricia Moore	Male	789 Lake Drive	+18576021493	pmoore@email.com	Tom Baker	+19028064671	Dr. Sarah Lee	Flu	06-Aug-2026	P004
	James Taylor	Male	456 River Road	+18786506280	jtaylor@email.com	Lisa Wong	+17328066683	Dr. Mark Davis	Cough	06-Aug-2026	P003
	Maria Garcia	Female	123 Sunset Blvd	+17402404851	mgarcia@email.com	John Smith	+19158529769	Dr. Emily Chen	Headache	06-Aug-2026	P002
	Robert Wilson	Female	890 Cedar Ave	+19495985566	rwilson@email.com	Sarah Connor	+17198825859	Dr. Alan Grant	Fever	06-Aug-2026	P001

Result: The cloud-based Hospital Information System was successfully developed and tested. Patient records were managed efficiently using cloud technology, successfully demonstrating the Software as a Service (SaaS) model.

Experiment 2: Create a Cloud-based Online Supermarket Application (SaaS)

Aim: To develop a cloud-based Online Supermarket application using a Cloud Service Provider for customer order management and item classification, demonstrating the Software as a Service (SaaS) model.

Tools / Platform

- Google App Sheet
- Google Sheets
- Google Drive
- Zoho creator
- Internet Connection

Procedure

1. Create a Google Sheet containing Customer Name, Address, Phone Number, Email, Item Name, Category, Quantity, Price, and Total Amount.
2. Open Google App Sheet and connect the Google Sheet.
3. Generate the supermarket application automatically.
4. Design customer registration and order entry forms.
5. Configure item categories such as Grocery, Dairy, Fruits, Vegetables, Bakery, and Beverages.
6. Test the application by placing sample customer orders.
7. Verify that customer orders are stored and updated automatically in the cloud database.
8. Publish the application for online access.

Application of Concepts

- Implemented the SaaS cloud model for online shopping.

- Stored customer and order details securely in cloud storage.
- Used cloud databases for centralized order management.
- Enabled real-time synchronization of customer orders.
- Allowed users to access the application from any internet-enabled device.

Analysis and Output

The Online Supermarket application successfully managed customer information, product selection, item classification, quantity, and billing details. Cloud synchronization ensured that order information was updated instantly. The application demonstrated the benefits of cloud computing, including scalability, availability, and centralized management.

 **Online SuperMarket**

 **Customer Information Form**

Customer Details

Customer ID

CUS-0001

Customer Name *

Enter Customer Name

Address *

Enter Address

Phone Number *

 Enter 10 Digit Mobile Number

Email Address *

 Enter Email Address

City

Enter City

Pincode

Enter Pincode

 Save

 Reset



Item Classification Form

Item Details

Item Code *

Item Name *

Category *

Sub Category

Unit Price (₹) *

Unit *

Stock Available *

Reorder Level

Description

Save

Reset

Items Report *

Save Changes

Remove Changes

		Item ID	Item Name	Item Code	Unit Price	Stock Available
		5	Green Leafy Spinach	ITM005	\$ 1.80	1200
		4	Free Range Eggs	ITM004	\$ 4.50	600
		3	Whole Milk	ITM003	\$ 3.75	2000
		2	Fresh Apples	ITM002	\$ 2.50	800
		1	Organic Bananas	ITM001	\$ 1.25	1500

Result: The cloud-based Online Supermarket application was successfully developed and tested. Customer orders and product information were managed efficiently, demonstrating the practical implementation of the SaaS cloud model.

Experiment 3: Installation and Configuration of Type-2 Hypervisor

Aim: To install a Type-2 Hypervisor and create a virtual machine with 3 CPU cores, 10 GB RAM, and a 10 GB virtual disk for demonstrating virtualization.

Tools / Platform

- VMware Workstation
- Windows/Linux Host Operating System
- Ubuntu Linux ISO Image
- Internet Connection

Procedure

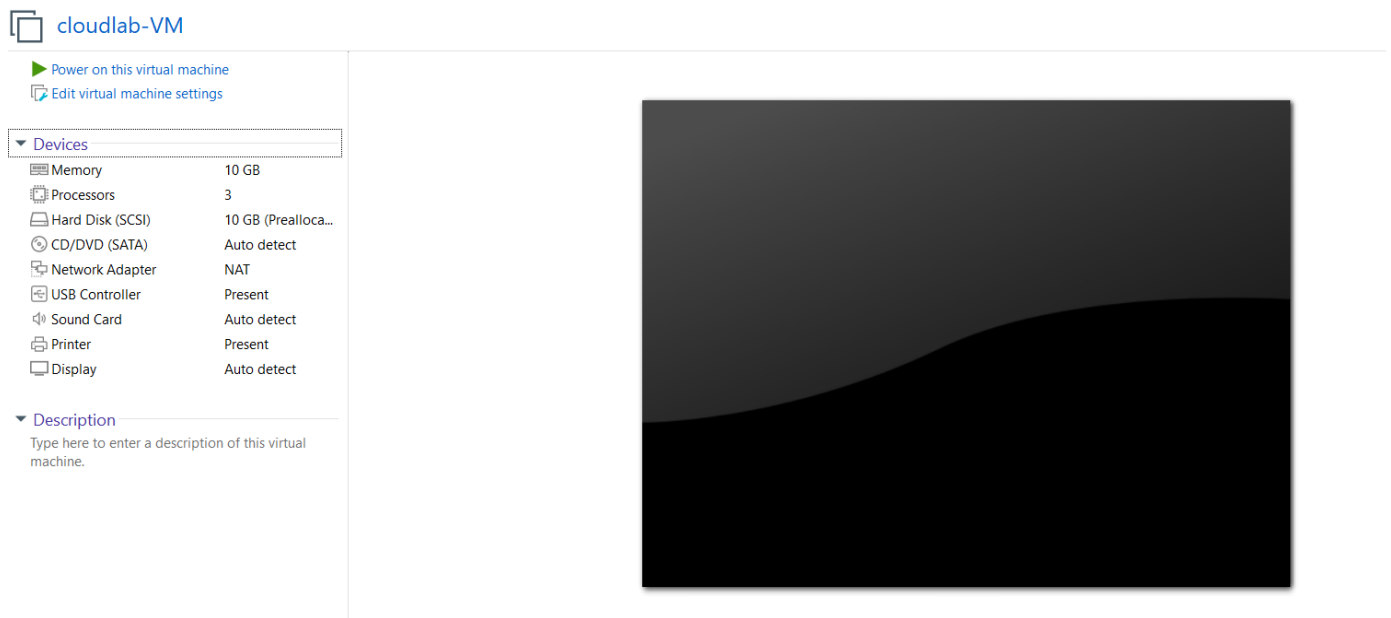
1. Download and install Oracle VM VirtualBox.
2. Download the Ubuntu Linux ISO image.
3. Open VirtualBox and create a new virtual machine.
4. Allocate 3 CPU cores, 10 GB RAM, and a 10 GB virtual hard disk.
5. Attach the Ubuntu ISO image to the virtual machine.
6. Install Ubuntu Linux as the guest operating system.
7. Boot the virtual machine and complete the operating system setup.
8. Verify the allocated CPU, RAM, and storage using Linux system commands.

Application of Concepts

- Demonstrated hardware virtualization using a Type-2 Hypervisor.
- Created an isolated guest operating system.
- Allocated virtual hardware resources to the guest machine.
- Enabled multiple operating systems to run on a single physical computer.
- Demonstrated resource sharing and system isolation.

Analysis and Output

The virtual machine was created successfully with the required configuration. Ubuntu Linux was installed and executed without affecting the host operating system. The experiment demonstrated efficient hardware utilization, virtualization, and independent execution of multiple operating systems.



Result

The Type-2 Hypervisor was successfully installed, and a virtual machine with 3 CPU cores, 10 GB RAM, and a 10 GB virtual disk was created and configured successfully, demonstrating virtualization technology.

Experiment 4: VM Snapshot Creation and Restoration

Aim: To create and restore a snapshot of a virtual machine using a Type-2 Hypervisor for demonstrating virtual machine backup and recovery.

Tools / Platform

- VMware Workstation
- Ubuntu Linux Virtual Machine
- Snapshot Manager

Procedure

1. Start the Ubuntu virtual machine.
2. Install additional software or modify system settings.
3. Shut down the virtual machine.
4. Open the Snapshot Manager in VirtualBox.
5. Create a snapshot and provide an appropriate name.
6. Restart the virtual machine and make additional system changes.
7. Restore the previously created snapshot.
8. Restart the virtual machine and verify that it has returned to its earlier state.

Application of Concepts

- Demonstrated virtual machine snapshot technology.
- Performed backup and recovery of a virtual machine.
- Restored the system to a previous working state.
- Illustrated rollback capability after software installation or configuration changes.

- Demonstrated the importance of snapshots for testing, maintenance, and disaster recovery.

Analysis and Output

The virtual machine snapshot was created and restored successfully. After restoration, all modifications made after the snapshot were removed, returning the virtual machine to its previous stable condition. The experiment highlighted the effectiveness of snapshots in protecting virtual environments and simplifying system recovery.

 VIRTUAL MACHINE

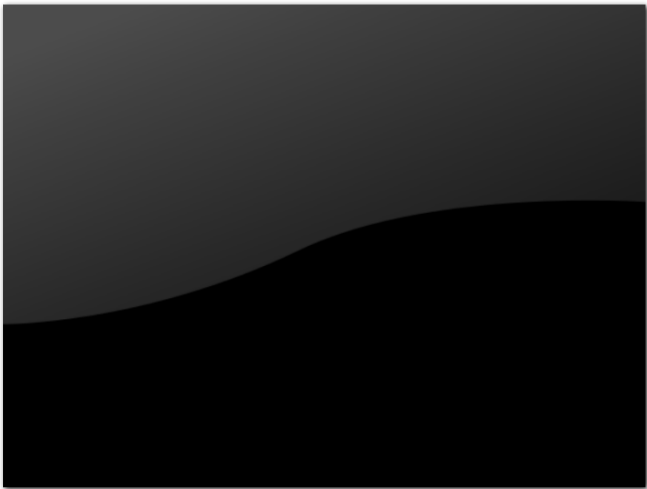
[Power on this virtual machine](#)
[Edit virtual machine settings](#)

▼ Devices

Memory	10 GB
Processors	3
Hard Disk (SCSI)	10 GB (Prealloca...
CD/DVD (SATA)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

▼ Description

Type here to enter a description of this virtual machine.



▼ Virtual Machine Details

State: Powered off
Snapshot: Snapshot 2
Configuration file: C:\Users\umesh\OneDrive\Documents\Virtual Machines\Clone of Ubuntu 64-bit (2)\Clone of Ubuntu 64-bit (2).vmx
Clone of: C:\Users\umesh\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 16.2.x virtual machine
Primary IP address: Network information is not available

Result: The snapshot was successfully created and restored, proving that virtualization technology supports efficient backup, rollback, and recovery of virtual machines without reinstalling the operating system.
